Rule-Constrained Fine-Tuning and Task Transfer in Large Language Models: Behavioral Evaluation and Causal Analysis of Attention Heads

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Abstract

Large language models can acquire domain-specific behavioral patterns through task fine-tuning, yet it remains unclear whether such patterns transfer to reasoning tasks with different surface forms and how fine-tuning changes the internal mechanisms associated with the learned behavior. This study uses Dou Dizhu as a structured rule-based decision environment to investigate task transfer and constraint-sensitive preference reallocation after fine-tuning on a rule-constrained mixed corpus. We construct a mixed fine-tuning corpus containing general instruction data and Dou Dizhu decision data, with explicit prohibitive constraints introduced into a subset of the decision examples.

To evaluate cross-task transfer, we develop ACL Bench, a 300-item benchmark that abstracts source-domain operations into three reasoning categories: Hierarchy Logic, Pattern Recognition, and Counter Logic. Across three base-model families, fine-tuned models achieve higher overall ACL Bench accuracy, while category-level gains vary substantially across models. These results indicate selective and model-dependent transfer rather than uniform improvement in general reasoning.

We further construct 60 controlled constraint-sensitive decision probes and introduce the best constraint delta margin, which measures whether an explicit prohibition shifts model preference from a forbidden action toward the strongest legal alternative. On Qwen2.5-7B, the mean score increases from 1.391 ± 1.379 in the Base model to 4.371 ± 0.766 after fine-tuning, indicating substantially stronger constraint-sensitive preference reallocation.

To examine the internal mechanisms associated with this behavioral change, we perform causal ablation of all 784 attention heads in both the Base and fine-tuned models and quantify the Base-to-LoRA change in causal contribution using ΔCE. The resulting causal effects are sparse and heterogeneous, with a limited subset of heads showing strongly amplified importance after fine-tuning. Joint Top-k ablation further shows that removing the highest-ranked heads produces substantially greater behavioral degradation than both globally random and layer-matched random controls across all tested intervention sizes. The non-monotonic Top-k effects additionally indicate that head contributions are not simply additive.

Overall, these findings suggest that fine-tuning can selectively reshape behavioral preferences and the causal contributions of specific internal components, while highlighting that preference-level adaptation should be distinguished from final-action compliance.

1 Introduction

Large language models (LLMs), built upon Transformer architectures and large-scale pretraining, can acquire new task behaviors through fine-tuning, allowing a pretrained model to adapt to specific domains, instructions, and decision requirements [15, 1]. Instruction tuning has further shown that behaviors learned from collections of supervised tasks can generalize beyond the exact tasks observed during training, improving performance on previously unseen task types [16, 3]. Parameter-efficient methods such as Low-Rank Adaptation (LoRA) have made this form of adaptation increasingly practical by enabling substantial behavioral changes without updating all model parameters [7]. These advances raise a broader question: when a model is fine-tuned to follow a structured decision policy in one domain, which aspects of the learned behavior remain specific to that domain, and which can transfer to reasoning tasks with different surface forms?

This question is particularly important for fine-tuning data that encode not only preferred actions but also explicit constraints on how decisions should change. In many realistic settings, successful decision making requires more than learning a fixed mapping from an input to a preferred output. A previously preferred action may become invalid after an additional rule, prohibition, or contextual condition is introduced, requiring the model to suppress that action and select an appropriate alternative. We refer to this form of supervision as *rule-constrained fine-tuning*. Compared with ordinary task fine-tuning, such examples explicitly train the model to revise its decision preference when the admissible action space changes. This makes rule-constrained training a useful setting for studying whether conditional decision patterns can transfer beyond their original domain.

Existing evidence from instruction tuning demonstrates that fine-tuned language models can generalize to unseen tasks [16, 3], but broad benchmark improvements alone do not reveal what is actually being transferred. Improvements may arise from task-specific knowledge, shared linguistic templates, general instruction-following ability, or reusable reasoning operations. A stricter study of task transfer therefore requires separating the *surface form* of a task from the *structural operations* needed to solve it. In particular, if a model trained in one decision domain improves on tasks that use different terminology and contexts but preserve operations such as hierarchical comparison, pattern identification, or conditional candidate selection, the resulting behavior provides evidence consistent with transfer at the level of structural operations.

To investigate this problem in a controlled setting, we use Dou Dizhu as a structured rule-based decision environment. The purpose of using Dou Dizhu is not to study game playing as an end in itself. Rather, the domain provides a compact environment containing explicit hierarchical relations, combinatorial patterns, legal and illegal actions, and conditional decision changes. We construct a mixed fine-tuning corpus consisting of general instruction data and Dou Dizhu decision data. In a subset of the decision examples, the action that would otherwise be preferred is explicitly prohibited, requiring the model to select a legal alternative. The resulting training setting allows us to study how a model adapts when the relationship between state and preferred action becomes conditional on an additional rule.

We study the resulting behavior from two complementary perspectives. First, we ask whether the fine-tuned decision patterns transfer to tasks that are structurally related to the source domain but differ substantially in surface form. To this end, we construct *ACL Bench*, a 300-item multiple-choice benchmark containing three reasoning categories: Hierarchy Logic, Pattern Recognition, and Counter Logic. These categories abstract recurring operations from the source decision environment while removing game-specific wording and scenarios. ACL Bench therefore serves as a controlled test of whether fine-tuning is associated with transfer at the level of reasoning structure rather than direct reproduction of domain-specific examples.

Second, we examine how fine-tuning changes the model's response to explicit constraints. Measuring only whether the probability of a prohibited action decreases is insufficient, because suppression of one action does not necessarily imply selection of an appropriate alternative. We therefore construct controlled constraint-sensitive decision probes and introduce the *best constraint delta margin*, a log-probability-based metric that measures how the relative preference between a forbidden action and the strongest legal alternative changes after a constraint is introduced. This formulation views constraint-sensitive decision adaptation as a problem of *preference reallocation*: the model should not merely suppress an invalid action, but should reorganize its decision distribution toward a valid alternative.

Behavioral evaluation alone, however, cannot determine how fine-tuning changes the internal mechanisms associated with the learned behavior. Mechanistic interpretability studies have shown that causal interventions and circuit-level analyses can be used to identify model components that are functionally involved in specific behaviors, rather than relying only on observed activation or attention patterns [9, 4, 5]. Previous work has also demonstrated that individual attention heads can implement behaviorally meaningful computations in Transformer models [10]. Motivated by this intervention-based perspective, we perform a complete attention-head causal ablation analysis on the fine-tuned model. For every attention head, we measure the change in the best constraint delta margin when that head is ablated and compare its contribution before and after fine-tuning.

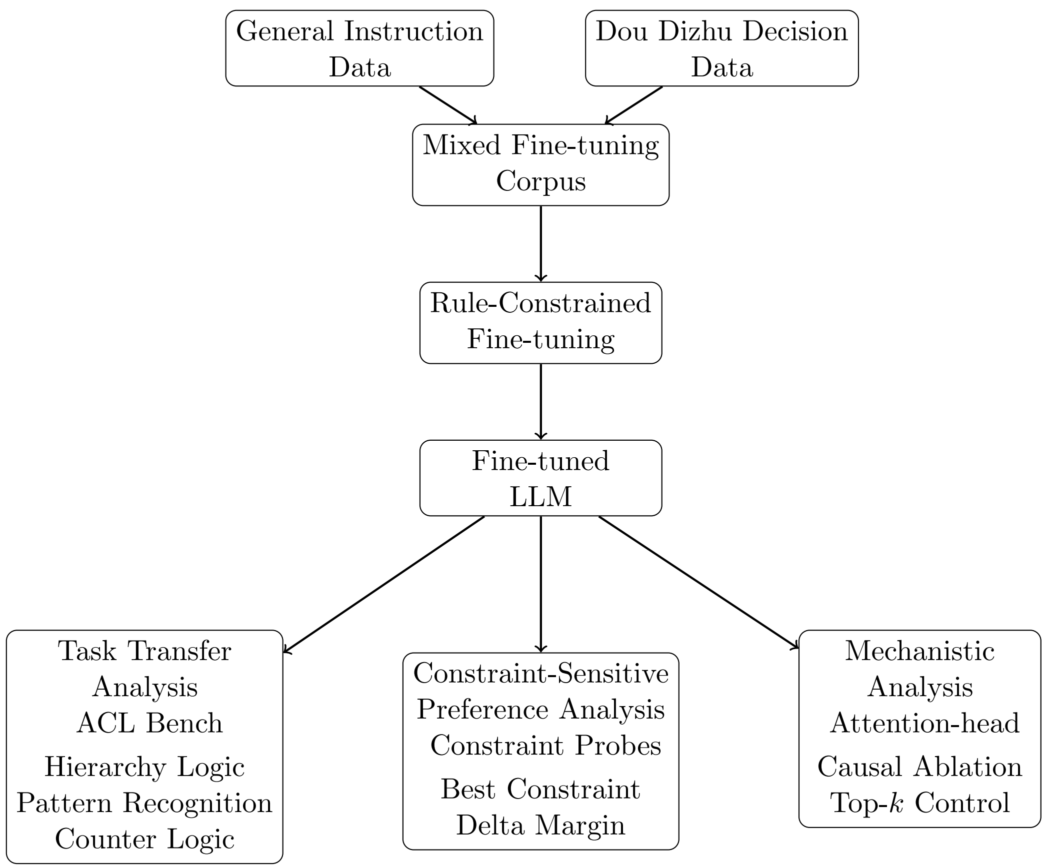


Figure 1: Overview of the proposed rule-constrained fine-tuning framework. The framework integrates structured domain fine-tuning, cross-task transfer evaluation, constraint-sensitive behavioral analysis, and causal investigation of internal model changes.

We further test whether the highest-ranked attention heads constitute a genuinely important functional subset rather than isolated single-head effects. Heads are ranked according to the increase in their causal contribution after fine-tuning, denoted as ΔCE. We then jointly ablate the Top-k ranked heads and compare the resulting behavioral degradation with two controls: globally random head sets and layer-matched random head sets. The latter preserves the layer distribution of the

Top-k intervention and therefore controls for the possibility that the observed effect is explained simply by removing heads from particular network depths.

Our experiments reveal three main patterns. First, models fine-tuned with the rule-constrained corpus exhibit cross-task transfer on ACL Bench across multiple base-model families, but the gains are strongly dependent on both model family and reasoning category. The resulting transfer is therefore selective rather than a uniform improvement in general reasoning. Second, fine-tuning substantially strengthens constraint-sensitive preference reallocation: after a preferred action becomes prohibited, the fine-tuned model more strongly shifts relative probability toward legal alternatives. Third, the causal contribution of attention heads changes in a sparse and heterogeneous manner. A limited subset of heads exhibits substantially amplified causal importance after fine-tuning, and joint ablation of the highest-ranked heads produces considerably larger behavioral degradation than either global-random or layer-matched random controls. At the same time, the Top-k effect is non-monotonic as k increases, indicating that attention-head contributions are not simply additive and may involve redundancy or nonlinear interaction.

These findings support a view of fine-tuning in which behavioral adaptation is accompanied by selective changes in the functional contribution of pre-existing computational mechanisms rather than uniform modification of the network. Importantly, the current causal analysis directly concerns constraint-sensitive decision behavior. Although ACL Bench provides independent evidence of cross-task transfer, the present experiments do not yet establish that the same attention heads causally mediate the transfer observed on ACL Bench. We therefore treat task-transfer evaluation and mechanistic analysis as complementary levels of evidence rather than assuming a direct causal connection between them.

The main contributions of this work are as follows:

- We formulate rule-constrained fine-tuning as a controlled setting for studying how conditional decision behavior learned in a structured source domain relates to reasoning performance on tasks with different surface forms.
- We introduce ACL Bench, a 300-item benchmark that abstracts source-domain decision structures into Hierarchy Logic, Pattern Recognition, and Counter Logic, enabling fine-grained evaluation of structurally related task transfer.
- We propose the best constraint delta margin and a controlled probe framework for measuring how model preference changes when an explicit constraint invalidates a previously preferred action.
- We conduct a complete attention-head causal ablation analysis and introduce a Base-to-LoRA causal-change measure, ΔCE, to characterize how fine-tuning redistributes functional dependence across attention heads.
- We validate the functional importance of high-ΔCE heads through Top-k joint ablation against both global-random and layer-matched random controls, providing evidence that the learned constraint-sensitive preference shift is associated with a sparse causal subset rather than on uniform changes across the attention system.

Together, these analyses connect cross-task behavioral evaluation, constraint-sensitive decision measurement, and causal intervention within a unified experimental framework for studying capability transfer after fine-tuning.

2 Related Work

2.1 Fine-Tuning and Cross-Task Generalization

A central motivation for instruction tuning is that supervision on a collection of tasks can improve performance beyond the exact training instances. Built upon Transformer architectures and large-scale pretraining, modern language models have demonstrated increasingly broad capabilities and adaptability across diverse tasks [15, 1]. Early large-scale studies demonstrated that converting heterogeneous supervised tasks into natural-language instructions enables language models to generalize to held-out tasks. FLAN showed that instruction-tuned language models can achieve substantially improved zero-shot performance on previously unseen tasks [16]. T0 similarly demonstrated that multitask prompted training can induce zero-shot task generalization across held-out datasets and task formulations [12]. Subsequent work further showed that increasing the number and diversity of instruction-tuning tasks, model scale, and chain-of-thought supervision can substantially improve generalization across a broad range of benchmarks [3].

These studies establish that fine-tuning can produce behavioral effects beyond the immediate training tasks. However, most instruction-tuning work evaluates generalization across heterogeneous benchmark collections in which the relationship between training and evaluation tasks is often defined at the dataset or task-family level. As a result, strong held-out performance does not by itself reveal which computational or reasoning structures have transferred. Improvements may reflect better interpretation of instructions, broader task exposure, shared linguistic patterns, or more reusable task-solving operations.

The present work focuses on a narrower form of transfer. Rather than training on a large collection of diverse tasks, we fine-tune models in a structured source domain and evaluate whether recurring decision operations transfer to tasks with different surface forms. In particular, ACL Bench removes Dou Dizhu-specific terminology while retaining abstract operations related to hierarchical comparison, pattern structure, and conditional candidate selection. Our objective is therefore not to demonstrate general zero-shot task competence, but to examine whether fine-tuning in one structured decision environment is associated with selective transfer to structurally related reasoning tasks. This perspective is related to recent efforts in evaluating broad language model capabilities through comprehensive benchmarks such as MMLU and BIG-Bench, which measure performance across diverse reasoning and knowledge tasks [6, 13]. In contrast, our benchmark is designed to isolate transfer of specific structural operations under controlled changes of surface form.

This distinction also motivates category-level analysis. If fine-tuning produced a uniform increase in general reasoning ability, improvements would be expected to occur broadly across target tasks. In contrast, selective improvements concentrated in structurally related categories would be more consistent with operation-level transfer. Our evaluation is designed to distinguish between these possibilities.

Parameter-efficient fine-tuning further makes this question practically relevant. Low-Rank Adaptation (LoRA) enables substantial model adaptation while updating only a small number of additional parameters [7]. Because LoRA can induce large behavioral changes without full-model retraining, understanding whether such changes remain source-specific or extend to related reasoning operations is important for both adaptation and interpretability.

2.2 Constraint Following and Rule-Aware Decision Making

Instruction following requires language models not only to generate plausible responses but also to satisfy explicit conditions imposed by the user or task. Recent benchmarks have therefore moved beyond general response quality toward direct evaluation of constraint satisfaction.

IFEval introduced a reproducible benchmark based on verifiable instructions, including requirements concerning response length, keywords, formatting, and other objectively checkable conditions [17]. FollowBench extended this direction by evaluating multiple categories of fine-grained constraints, including content, situation, style, format, and example-related requirements, and by increasing difficulty through progressively more constrained instructions [8]. These benchmarks demonstrate that constraint following is a distinct capability and that even strong language models can fail when instructions contain multiple or increasingly specific requirements.

The constraints studied in the present work differ from many output-level instruction constraints in an important way. Our constraint examples modify the *decision space* rather than merely specifying properties of the final response. A model is first placed in a state where one action would normally be preferred. An additional rule then explicitly prohibits that action, requiring the model to revise its preference and select a legal alternative.

This distinction can be expressed as the difference between satisfying an output condition,

$$y \in \mathcal{C},$$

and adapting a decision policy after the admissible action set changes,

$$a^* = \arg\max_{a \in \mathcal{A}} P(a \mid x) \quad \longrightarrow \quad a^c = \arg\max_{a \in \mathcal{A} \setminus \mathcal{F}_c} P(a \mid x, c),$$

where $\mathcal{F}_c$ denotes the set of actions forbidden by constraint c.

This formulation motivates our emphasis on *preference reallocation*. A model may satisfy a prohibition by decreasing the probability of the forbidden action, yet still fail to redirect probability toward an appropriate legal alternative. Measuring only forbidden-action suppression therefore provides an incomplete characterization of constraint-sensitive decision making.

Our best constraint delta margin is designed to capture this distinction by measuring the relative preference shift between the prohibited action and the strongest legal alternative. In this sense, our work complements existing constraint-following benchmarks: rather than primarily asking whether the generated response satisfies a set of surface-level requirements, we study how fine-tuning changes a model's internal action preference when a previously preferred decision becomes invalid.

The rule-constrained setting also provides a controlled way to study conditional decision policies. Standard supervised examples often reinforce a mapping of the form

$$x \to a^*,$$

whereas constraint-injected examples introduce supervision of the form

$$(x, c) \to a^c.$$

The same underlying state can therefore require different actions depending on an additional rule. This property makes the setting particularly suitable for examining whether learned conditional decision patterns extend to other forms of structured reasoning.

2.3 Mechanistic Interpretability of Fine-Tuned Language Models

Behavioral evaluation can establish that fine-tuning changes model performance, but it cannot determine which internal computations become more or less important after adaptation. Mechanistic interpretability addresses this problem by using interventions on internal model components to identify computations that are functionally involved in specific behaviors. Circuit-level analyses further investigate how Transformer components such as attention heads contribute to specific computational patterns [5, 9, 4].

Studies of Transformer models have shown that individual attention heads can implement recognizable computational functions. Olsson et al. identified induction heads that support sequence-completion behavior and provided causal evidence for their functional role in smaller attention-only models [10]. This line of work illustrates that attention heads can sometimes serve as meaningful units for analyzing model computation rather than being treated only as components of an undifferentiated attention layer.

Intervention-based methods further distinguish causal importance from observational patterns. Meng et al. used causal interventions to localize internal computations responsible for factual associations in autoregressive language models, showing that targeted perturbations can reveal components that are decisive for model predictions [9]. Automated Circuit Discovery subsequently formalized circuit identification as an intervention-based process in which model components and connections are evaluated according to their contribution to a specified behavioral metric [4].

This distinction is particularly relevant for attention analysis. High attention weights or strong correlations between attention patterns and outputs do not necessarily imply that the corresponding component is causally necessary for the behavior. A component may participate in a distributed computation without exhibiting a simple observable correlation with the final prediction. For this reason, the present study uses causal head ablation as the primary measure of functional importance and treats attention-based correlation screening as exploratory rather than definitive evidence.

More recent work has begun to investigate how fine-tuning changes model mechanisms directly. Prakash et al. studied entity tracking before and after fine-tuning and found that fine-tuned models largely reused a pre-existing circuit, with improved performance arising from enhancement of mechanisms already present in the base model [11]. This finding supports the possibility that fine-tuning can amplify or reconfigure existing computations rather than constructing entirely new mechanisms.

Related evidence has also shown that fine-tuning can modify localized circuit components in more complex ways. Chhabra et al. analyzed model mechanisms under task-specific and corrupted fine-tuning and reported amplification of task-relevant mechanisms, localized corruption, and subsequent recovery of previously used mechanisms during retraining [2]. These results further suggest that fine-tuning can alter the relative functional importance of existing components rather than uniformly changing the network.

Our work builds on this emerging line of research but differs in both the target behavior and the causal comparison. We study a rule-constrained decision behavior in which the model must revise its preferred action under an explicit prohibition. Rather than identifying a complete predefined circuit, we perform a complete single-head intervention scan and compare each head's behavioral contribution before and after fine-tuning. We define

$$\Delta CE(l,h) = CE_{\text{LoRA}}(l,h) - CE_{\text{Base}}(l,h)$$

to quantify how the causal importance of each attention head changes after adaptation.

We then extend single-component analysis with joint Top-k interventions. The highest-ranked heads are ablated collectively and compared against both globally random and layer-matched random head sets. This control design allows us to test whether large single-head causal changes identify a functionally distinctive subset, rather than merely reflecting generic sensitivity to multi-head removal or the concentration of selected heads within particular layers.

Thus, the present study connects three lines of prior research: cross-task generalization after fine-tuning, explicit constraint following, and mechanistic analysis of fine-tuned models. Our contribution lies in bringing these perspectives into a single controlled setting, where structural task transfer, constraint-sensitive preference reallocation, and changes in attention-head causal dependence can be studied together.

3 Method

Our study investigates rule-constrained fine-tuning from three complementary levels: source-domain training, cross-task behavioral transfer, and internal causal dependence. We first construct a mixed fine-tuning corpus in which a subset of structured decision examples contains explicit prohibitive constraints. We then evaluate cross-task transfer using ACL Bench, a benchmark designed to preserve structural operations from the source domain while changing their surface form. To characterize the learned constraint-sensitive preference reallocation more directly, we construct controlled decision probes and introduce the best constraint delta margin. Finally, we perform complete single-head causal ablation and Top-k joint interventions to examine how fine-tuning changes the functional importance of individual attention heads.

3.1 Dataset Construction

3.1.1 Mixed Fine-Tuning Corpus

We use Dou Dizhu as a structured rule-based decision environment. The purpose of the source domain is not to optimize game-playing performance itself, but to provide repeated examples of hierarchical comparison, structured pattern recognition, legal-action selection, and conditional decision adjustment.

The final fine-tuning corpus contains 55,900 instruction–response examples, consisting of 50,400 general instruction examples and 5,500 Dou Dizhu decision examples. The general instruction data are included to preserve broad instruction-following behavior, while the domain-specific examples expose the model to the structured decision operations of interest.

All fine-tuned models are adapted using Low-Rank Adaptation (LoRA) [7], allowing task-specific behavior to be learned while keeping the pretrained model parameters frozen.

3.1.2 Ordinary and Rule-Constrained Decision Examples

We distinguish between ordinary decision examples and rule-constrained decision examples.

An ordinary decision example can be represented as

$$x \to a^*,$$

where x denotes the current decision state and a^* is the preferred legal action under the original rules.

For the rule-constrained data, an additional condition c is introduced that explicitly prohibits an action that would otherwise be preferred. The resulting example takes the form

$$(x, c) \to a^c,$$

where a^c is a legal alternative under the modified action space.

The final rule-constrained corpus contains 851 constraint-injected instructions. Among these examples, 646 require a change in the target output after the additional constraint is introduced.

This construction exposes the model to cases in which a previously preferred action becomes unavailable and the decision policy must therefore be revised rather than simply reproduced.

Conceptually, the supervision changes from

$$a^* = \arg\max_{a \in \mathcal{A}} P(a \mid x)$$

to

$$a^c = \arg\max_{a \in \mathcal{A} \setminus \mathcal{F}_c} P(a \mid x, c),$$

where $\mathcal{A}$ denotes the original candidate-action set and $\mathcal{F}_c$ denotes the set of actions prohibited by constraint c.

The model trained on the rule-constrained corpus is referred to as **LoRA-v2** throughout the paper. An ordinary version of the mixed fine-tuning corpus without the special constraint intervention is retained as LoRA-v1. However, the primary results in the present study focus on the Base and LoRA-v2 comparison. A strictly matched LoRA-v1 versus LoRA-v2 experiment is reserved for isolating the incremental contribution of the constraint-injected examples and is not included in the current main results.

3.2 ACL Bench Construction

To evaluate whether behavioral patterns acquired in the source domain transfer to tasks with different surface forms, we construct ACL Bench.

ACL Bench contains 300 four-choice multiple-choice questions divided equally into three reasoning categories:

- **Hierarchy Logic**: reasoning over explicitly ordered hierarchical relations, including comparison and rule-dependent access or override relations;
- **Pattern Recognition**: identifying, extending, or distinguishing structured patterns, including sequence completion and outlier recognition;
- **Counter Logic**: selecting the minimum candidate that strictly exceeds a specified threshold, including cases in which no candidate satisfies the condition.

Each category contains 100 questions.

The benchmark is designed to preserve abstract operations that recur in the source decision environment while removing direct dependence on Dou Dizhu-specific terminology. Consequently, successful performance does not require knowledge of card-game rules, card names, or source-domain output templates.

The three categories capture different structural relationships between the source and target tasks. Hierarchy Logic abstracts ordered rank relations, Pattern Recognition captures recurring structural regularities, and Counter Logic abstracts conditional selection relative to a comparison threshold.

The benchmark generation procedure uses a fixed random seed of 42. Answer options are shuffled during item construction, and the complete benchmark is additionally shuffled before export.

Several quality-control procedures are applied during benchmark construction. Hierarchy Logic items are checked for subject–relation alignment, Pattern Recognition items are checked to prevent duplicated answer options, and Counter Logic prompts are revised to ensure that the comparison condition is expressed consistently. Additional manual and programmatic checks are used to remove typographical errors, rare or unintended lexical artifacts, and context–answer mismatches.

For a model M, category-level accuracy is defined as

$$\mathrm{Acc}_c(M) = \frac{1}{N_c} \sum_{i=1}^{N_c} \mathbb{I}\left[\hat{y}_i^M = y_i\right],$$

where $N_c = 100$ for each category, $\hat{y}_i^M$ is the answer predicted by model M, and y_i is the correct answer.

Overall ACL Bench accuracy is computed over all 300 questions:

$$\mathrm{Acc}_{\mathrm{ACL}}(M) = \frac{1}{300} \sum_{i=1}^{300} \mathbb{I}\left[\hat{y}_i^M = y_i\right].$$

The purpose of ACL Bench is therefore not to measure unrestricted general reasoning ability, but to test whether fine-tuning is associated with transfer across tasks that share selected structural operations with the source domain.

3.3 Constraint-Sensitive Preference-Shift Metric

ACL Bench evaluates cross-task transfer, but does not directly measure whether the model has learned to revise its decision preference when an explicit prohibition is introduced.

To examine this behavior, we construct 60 controlled constraint-sensitive decision probes covering six decision scenarios.

Each probe contains two closely matched conditions:

1. an unconstrained condition containing the original decision state;
2. a constrained condition in which an additional rule explicitly prohibits an otherwise preferred action.

For probe i, let f_i denote the forbidden action and let $\mathcal{L}_i$ denote the set of legal alternatives.

For a model M, we define the candidate-level log-probability score of action a under context x as

$$s_M(a \mid x) = \log P_M(a \mid x).$$

We first compute the relative margin between the strongest legal alternative and the forbidden action in the unconstrained condition:

$$m_i^{\mathrm{clean}} = \max_{a \in \mathcal{L}_i} s_M(a \mid x_i) - s_M(f_i \mid x_i).$$

After the prohibitive constraint c_i is introduced, the corresponding margin becomes

$$m_i^{\mathrm{constraint}} = \max_{a \in \mathcal{L}_i} s_M(a \mid x_i, c_i) - s_M(f_i \mid x_i, c_i).$$

We define the **best constraint delta margin** for probe i as

$$\mathrm{BCDM}_i = m_i^{\mathrm{constraint}} - m_i^{\mathrm{clean}}.$$

The model-level preference-shift score is the mean across all $N = 60$ probes:

$$S(M) = \frac{1}{N}\sum_{i=1}^{N} \mathrm{BCDM}_i.$$

A positive value indicates that introducing the constraint shifts relative model preference away from the forbidden action and toward the strongest legal alternative.

This metric differs from simple forbidden-action suppression. A model may reduce the probability of a prohibited action without increasing the probability of an appropriate legal response. By explicitly comparing the forbidden action with the strongest legal alternative, the best constraint delta margin measures *preference reallocation* rather than suppression alone.

3.4 Attention-Head Causal Ablation

Behavioral differences between the Base and fine-tuned models do not reveal which internal components become more or less important after adaptation. We therefore perform causal intervention at the level of individual attention heads.

The primary mechanistic analysis is conducted on Qwen2.5-7B. The model contains 28 Transformer layers and 28 attention heads per layer, resulting in

$$28 \times 28 = 784$$

attention heads.

For each head (l, h), we perform single-head zero ablation during the forward pass while leaving the remaining attention heads and model components unchanged. The resulting model is then reevaluated on the same 60 constraint-sensitive probes using the best constraint delta margin.

Let

$$S_{\text{original}}(M)$$

denote the preference-shift score of model M without intervention, and let

$$S_{\text{ablated}}(M, l, h)$$

denote the score after ablating attention head (l, h).

We define the causal effect of the head as

$$CE_M(l, h) = S_{\text{original}}(M) - S_{\text{ablated}}(M, l, h).$$

A positive value indicates that removing the head decreases constraint-sensitive preference shift, implying that the head makes a positive functional contribution to the measured behavior.

A negative value indicates that removing the head increases the score, suggesting that the head does not positively support the measured behavior under the current metric.

To quantify how fine-tuning changes the causal importance of each head, we define

$$\Delta CE(l, h) = CE_{\text{LoRA}}(l, h) - CE_{\text{Base}}(l, h).$$

A large positive ΔCE indicates that the model becomes more dependent on that attention head for constraint-sensitive preference shift after fine-tuning.

The complete procedure is applied independently to all 784 attention heads in both the Base and LoRA-v2 models, yielding 784 single-head interventions per model and 1,568 intervention measurements in total.

This complete scan allows us to analyze both individual high-impact heads and the global distribution of causal changes across layers and attention-head indices.

3.5 Top-k Causal Validation

Large single-head causal effects do not necessarily imply that the highest-ranked heads form a collectively important functional subset. We therefore perform joint Top-k ablation based on the ΔCE ranking.

Let

$$\mathcal{H}_k = \mathrm{TopK}_{(l,h)} \left[\Delta CE(l,h)\right]$$

denote the set containing the k attention heads with the largest positive Base-to-LoRA causal changes.

We evaluate three intervention sizes:

$$k \in \{5, 10, 27\}.$$

The $k = 27$ setting additionally corresponds to the complete set of heads with $\Delta CE > 1$ in the single-head causal ranking.

For each value of k, all heads in $\mathcal{H}_k$ are ablated simultaneously and the preference-shift score is recomputed.

The resulting behavioral degradation is defined as

$$D(\mathcal{H}_k) = S_{\mathrm{original}} - S_{\mathrm{ablated}}(\mathcal{H}_k).$$

A larger value indicates greater functional dependence on the removed head set.

To determine whether large Top-k effects are specific to the ΔCE ranking rather than a generic consequence of removing multiple attention heads, we construct two control conditions.

3.5.1 Global Random Control

The **Global Random** control selects k attention heads from the complete set of 784 heads without conditioning on layer position.

This control tests whether removing an arbitrary set of k heads is sufficient to produce degradation comparable to the Top-k intervention.

3.5.2 depth-bin-matched random Control

The **Matched Random** control preserves the layer-bin distribution of the corresponding Top-k set.

We divide the 28 Transformer layers into four depth ranges:

$$[0, 6], \quad [7, 13], \quad [14, 20], \quad [21, 27].$$

For each Top-k set, the matched control randomly samples the same number of heads from each corresponding depth range. This controls for coarse layer-depth distribution while preserving randomness within each range.

This provides a stricter comparison because it controls for the possibility that the Top-k effect is explained simply by ablating heads from particularly sensitive network depths.

The three intervention conditions are therefore

$$\text{Top-}k, \qquad \text{Global Random}, \qquad \text{Matched Random}.$$

For both random-control conditions, five independently sampled control sets are generated using random seeds 0, 1, 2, 3, and 4. For Global Random, the heads are sampled from the complete set of 784 attention heads. For Matched Random, sampling is performed independently within each layer while preserving the exact layer-count distribution of the corresponding Top-k set. The reported control effect for each value of k is the mean behavioral degradation across the five runs.

For both random-control conditions, we generate five independent control sets using random seeds 0, 1, 2, 3, and 4. The reported random-control effect for each value of k is the mean behavioral degradation across these five runs.

If the ΔCE ranking identifies a functionally distinctive subset, Top-k ablation should produce greater behavioral degradation than both control conditions.

Importantly, the Top-k intervention is treated as a joint causal test rather than as the sum of individual-head causal effects. Attention heads may exhibit redundancy, compensation, or nonlinear interaction, and therefore in general

$$CE(h_1, \ldots, h_k) \neq \sum_{i=1}^{k} CE(h_i).$$

The joint-ablation experiment is designed specifically to test the collective functional importance of the selected subset without assuming additive head contributions.

4 Experimental Setup

4.1 Base Models

We evaluate rule-constrained fine-tuning on three open-weight language models from different model families, including models derived from the LLaMA family [14]:

- **Qwen2.5-7B**;
- **Llama3.1-8B**;
- **DeepSeek-R1-Distill-Llama-8B**.

For each model family, the corresponding pretrained or instruction-tuned checkpoint before task-specific adaptation is referred to as the **Base** model. The model obtained after fine-tuning on the rule-constrained mixed corpus is referred to as **LoRA-v2**.

The use of multiple model families allows us to examine whether the observed transfer effect is consistent across different pretrained representations rather than being specific to a single architecture.

The primary mechanistic analysis is conducted on Qwen2.5-7B because the complete single-head causal scan and subsequent Top-k joint-ablation experiments were performed on this model.

4.2 Fine-Tuning Configuration

All task-specific adaptation experiments use Low-Rank Adaptation (LoRA) [7]. The pretrained model parameters remain frozen while low-rank trainable modules are inserted into selected projection layers.

Table 1: Main LoRA fine-tuning configuration.

Hyperparameter	Value
LoRA rank	16
LoRA alpha	32
LoRA dropout	0.05
Learning rate	1×10^{-4}
Training epochs	3
Per-device batch size	4
Maximum sequence length	2048
Learning-rate scheduler	Cosine
Warmup ratio	0.1
Gradient checkpointing	Enabled

Table 2: Recorded training statistics for the primary fine-tuned models.

Model	Training Steps	Final Recorded Loss	Training Time
Qwen2.5-7B	10,482	1.3892	11.04 h
Llama3.1-8B	20,964	1.1570	10.71 h
DeepSeek-R1-Distill-Llama-8B	20,964	1.0545	14.34 h

The main LoRA configuration is summarized in Table 1.

LoRA modules are applied to the attention projections

$$q_{\text{proj}}, \quad k_{\text{proj}}, \quad v_{\text{proj}}, \quad o_{\text{proj}},$$

as well as the feed-forward projections

$$\text{gate}_{\text{proj}}, \quad \text{up}_{\text{proj}}, \quad \text{down}_{\text{proj}}.$$

Thus, the adaptation is not restricted to the attention subsystem, even though the mechanistic analysis in this study focuses specifically on attention-head causal contributions.

The same overall fine-tuning framework is used across the evaluated model families to make the resulting behavioral comparisons as consistent as possible.

4.3 Training Runs

The principal fine-tuned checkpoints used in the current study are obtained from the rule-constrained mixed corpus described in Section 3.1.

Table 2 summarizes the recorded training statistics for the main LoRA-v2 models.

For Qwen2.5-7B, the recorded aggregate training loss for LoRA-v2 is 1.6312. The corresponding fine-tuned checkpoints are used for the behavioral evaluations described below.

Because the present study primarily concerns post-fine-tuning behavior and causal dependence rather than optimization dynamics, training loss is reported as an implementation statistic rather than as a central evaluation metric.

4.4 ACL Bench Evaluation

Cross-task transfer is evaluated using the 300-item ACL Bench introduced in Section 3.2.

Each model is evaluated on the same set of four-choice questions, consisting of:

- 100 Hierarchy Logic questions;
- 100 Pattern Recognition questions;
- 100 Counter Logic questions.

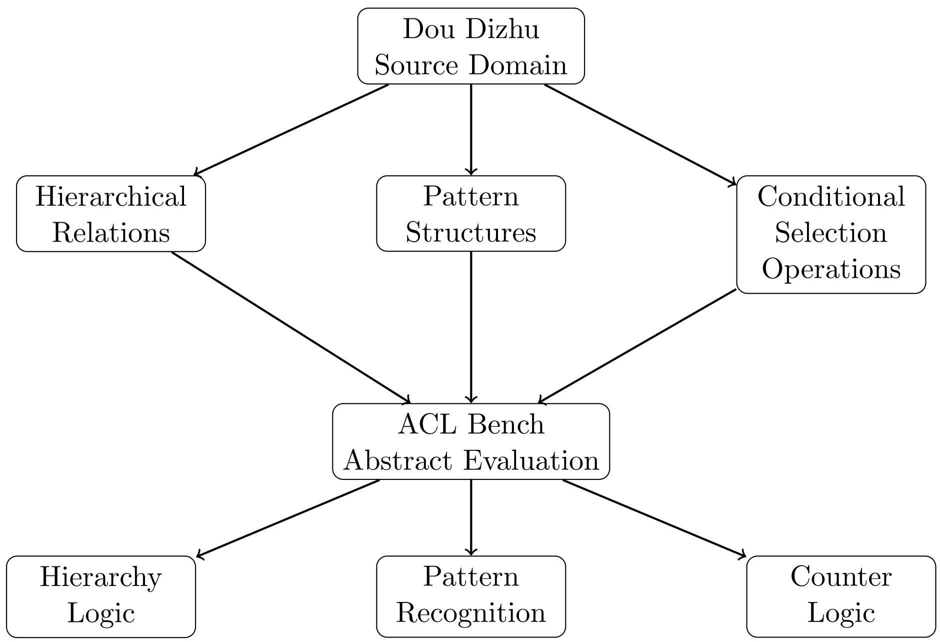


Figure 2: Construction of ACL Bench by abstracting Dou Dizhu decision operations into transferable reasoning categories. The benchmark removes domain-specific surface forms while preserving underlying decision operations.

We report both overall accuracy and category-level accuracy.

For each model family, performance is compared between the Base model and its LoRA-v2 counterpart. This paired design allows us to examine how task-specific fine-tuning changes performance relative to the same underlying model family.

The primary ACL analysis focuses on the direction and magnitude of Base-to-LoRA performance changes both overall and within individual reasoning categories.

Because category-specific transfer may differ substantially even when overall accuracy improves, category-level results are treated as a central part of the evaluation rather than as supplementary statistics.

4.5 Constraint-Sensitive Probe Evaluation

Constraint-responsive behavior is evaluated using 60 controlled decision probes spanning six decision scenarios.

The probe evaluation uses a fixed random seed of 42, FP16 inference, SDPA attention, and a batch size of 1.

Each probe contains a clean condition and a corresponding constrained condition in which an otherwise preferred action is explicitly prohibited.

All probes are evaluated using the best constraint delta margin defined in Section 3.3.

The primary comparison is between Qwen2.5-7B Base and Qwen2.5-7B LoRA-v2.

For each model, we report the mean behavioral score across all 60 probes:

$$S(M) = \frac{1}{60}\sum_{i=1}^{60} \mathrm{BCDM}_i.$$

We additionally summarize the score by scenario to examine whether the fine-tuning effect is consistent across different forms of constrained decision making.

The six probe scenarios are:

- `bomb_9999`;
- `pair_88`;
- `real_pair_QQ_landlord`;
- `rocket_BR`;
- `single_A_follow`;
- `straight_34567`.

These scenarios are used as controlled probes rather than as an independent benchmark. Their purpose is to measure how the model reallocates preference after an explicit prohibition is introduced.

4.6 Single-Head Causal Analysis

The complete attention-head causal analysis is performed on Qwen2.5-7B.

The model contains 28 Transformer layers with 28 attention heads per layer, yielding

$$28 \times 28 = 784$$

individual attention heads.

Each head is ablated independently and the model is reevaluated on the same 60 constraint-sensitive probes.

The complete scan is performed for both the Base and LoRA-v2 models, resulting in

$$784 \times 2 = 1568$$

single-head intervention evaluations.

For every head (l, h), we compute

$$CE_{\mathrm{Base}}(l, h), \qquad CE_{\mathrm{LoRA}}(l, h),$$

and their difference

$$\Delta CE(l, h) = CE_{\mathrm{LoRA}}(l, h) - CE_{\mathrm{Base}}(l, h).$$

The resulting 28×28 causal maps are used to examine the distribution of head-level causal contribution across both layer depth and head index.

Heads are then ranked according to ΔCE for subsequent joint-ablation analysis.

4.7 Top-k Joint-Ablation Evaluation

To validate whether the highest-ranked heads form a functionally distinctive subset, we perform joint ablation at three intervention sizes:

$$k \in \{5, 10, 27\}.$$

For each value of k, three intervention conditions are compared:

1. **Top-k**: the k heads with the largest positive ΔCE values;
2. **Global Random**: k heads selected from the full set of 784 attention heads;
3. **Matched Random**: k heads selected while preserving the layer-count distribution of the corresponding Top-k set.

The joint-ablation evaluation uses the same 60 constraint-sensitive probes with a fixed evaluation seed of 42, FP16 inference, SDPA attention, and a batch size of 1. For both random-control conditions, five independently sampled control sets are generated using random seeds 0, 1, 2, 3, and 4, and the reported random-control values are averaged across these five runs.

The joint-ablation evaluation uses the same 60 constraint-sensitive probes with a fixed evaluation seed of 42, FP16 inference, SDPA attention, and a batch size of 1.

The comparison is designed to distinguish three possible explanations for joint-ablation effects.

If Top-k and Global Random produce similar degradation, then the effect may reflect generic sensitivity to removing multiple attention heads.

If Top-k exceeds Global Random but is similar to Matched Random, the effect may be explained primarily by the network depths at which the selected heads are located.

If Top-k produces larger degradation than both controls, this provides stronger evidence that the identity of the selected high-ΔCE heads is functionally relevant to the learned constraint-sensitive preference shift.

No additive assumption is imposed on the joint interventions. The Top-k experiment measures the causal effect of ablating a set of heads simultaneously rather than summing their individual causal effects.

4.8 Implementation Environment

The primary fine-tuning and mechanistic experiments are conducted in a Windows Subsystem for Linux (WSL) environment on a workstation equipped with an NVIDIA RTX A4500 GPU with 20 GB of GPU memory.

Mixed-precision inference is used in the mechanistic evaluation pipeline where applicable, and gradient checkpointing is enabled during LoRA fine-tuning to reduce memory consumption.

The mechanistic evaluation is performed with a batch size of 1 for the constraint-sensitive probe analysis, allowing head-level interventions to be applied consistently across individual examples.

The current experimental design separates broad behavioral evaluation from detailed mechanistic analysis: cross-task transfer is evaluated across three model families, while complete attention-head causal analysis is concentrated on Qwen2.5-7B.

5 Results

This section reports the experimental results at three levels. We first evaluate cross-task transfer on ACL Bench across three model families. We then examine constraint-sensitive preference reallocation using the controlled behavioral probes. Finally, we analyze how fine-tuning changes the causal contribution of individual attention heads and validate the highest-ranked heads through joint Top-k ablation against two random-control conditions.

5.1 Task-Transfer Results

5.1.1 Overall ACL Bench Performance

Table 3 reports accuracy separately for Counter Logic, Hierarchy Logic, and Pattern Recognition.

Table 3: Category-level ACL Bench accuracy across three reasoning categories. Values are reported as percentages.

Model	Setting	Counter	Hierarchy	Pattern
DeepSeek-R1-Distill-Llama-8B	Base	30	27	27
	LoRA-v2	43	39	32
Qwen2.5-7B	Base	38	34	34
	LoRA-v2	24	70	68
Llama3.1-8B	Base	39	62	34
	LoRA-v2	41	65	70

All three model families achieve higher overall ACL Bench accuracy after fine-tuning.

DeepSeek-R1-Distill-Llama-8B improves from 28.0% to 38.0%, corresponding to a gain of 10.0 percentage points. Qwen2.5-7B exhibits the largest overall improvement, increasing from 35.3% to 54.0%, while Llama3.1-8B increases from 45.0% to 58.7%.

The consistency of the overall direction across three model families indicates that the behavioral changes induced by the rule-constrained fine-tuning corpus are not limited to a single pretrained model.

However, the category-level results reveal that these gains are highly non-uniform.

5.1.2 Category-Level Transfer

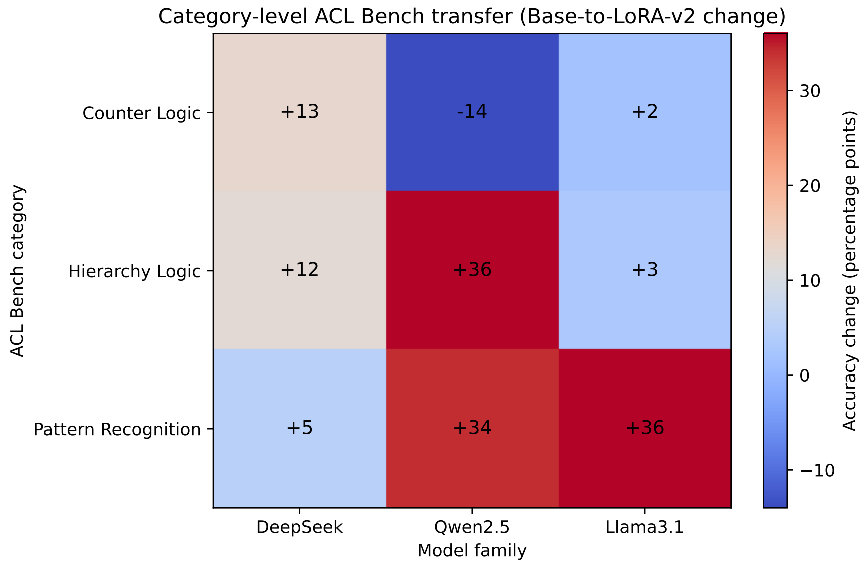
The category-level transfer pattern is summarized in Figure 3.

DeepSeek-R1-Distill-Llama-8B shows improvements across all three categories. Counter Logic increases from 30% to 43%, Hierarchy Logic from 27% to 39%, and Pattern Recognition from 27% to 32%. The improvement is therefore distributed across categories, although its magnitude varies.

Qwen2.5-7B exhibits a substantially different transfer profile. Hierarchy Logic increases from 34% to 70%, and Pattern Recognition increases from 34% to 68%. In contrast, Counter Logic decreases from 38% to 24%.

Llama3.1-8B shows only limited changes in Counter Logic and Hierarchy Logic, increasing from 39% to 41% and from 62% to 65%, respectively. Its largest improvement occurs in Pattern Recognition, which increases from 34% to 70%.

These category-level differences show that the overall gains cannot be interpreted as a uniform increase in general reasoning ability.



Model	Setting	Counter	Hierarchy	Pattern
DeepSeek	Base	30	27	27
	LoRA-v2	43	39	32
Qwen2.5	Base	38	34	34
	LoRA-v2	24	70	68
Llama3.1	Base	39	62	34
	LoRA-v2	41	65	70

Figure 3: Category-level ACL Bench transfer after rule-constrained fine-tuning. The heatmap shows Base-to-LoRA-v2 accuracy changes, while the table provides the corresponding category-level scores. The results reveal selective and model-dependent transfer rather than uniform improvement across reasoning categories.

Instead, the observed transfer is selective and strongly dependent on both the target reasoning category and the underlying base model.

In particular, Qwen2.5-7B and Llama3.1-8B obtain large improvements in Pattern Recognition, whereas their Counter Logic behavior differs substantially. Qwen2.5-7B improves strongly in Hierarchy Logic but degrades in Counter Logic, while Llama3.1-8B remains comparatively stable in both categories.

The ACL results therefore support a characterization of the fine-tuning effect as structure-dependent and model-dependent cross-task transfer rather than broad, homogeneous reasoning enhancement.

5.2 Constraint-Sensitive Preference-Shift Results

ACL Bench evaluates transfer outside the source domain, but it does not directly measure whether the fine-tuned model has learned to revise its decision preference when an additional rule prohibits an otherwise preferred action.

We therefore evaluate Qwen2.5-7B Base and LoRA-v2 on the 60 controlled constraint-sensitive probes using the best constraint delta margin.

Table 4: Best constraint delta margin on the 60 constraint-sensitive probes.

Model	Mean	Standard Deviation
Qwen2.5-7B Base	1.391	1.379
Qwen2.5-7B LoRA-v2	4.371	0.766

Table 5: Scenario-level best constraint delta margin for Qwen2.5-7B. Each scenario contains 10 probes.

Scenario	Base	LoRA-v2	Δ
`bomb_9999`	1.898	4.180	+2.282
`pair_88`	0.309	5.284	+4.975
`real_pair_QQ_landlord`	0.155	4.107	+3.952
`rocket_BR`	1.115	5.427	+4.312
`single_A_follow`	0.700	3.252	+2.552
`straight_34567`	4.168	3.977	-0.191

5.2.1 Overall Preference-Reallocation Effect

Table 4 summarizes the overall preference-shift scores.

The Base model obtains a mean best constraint delta margin of

$$1.390711 \pm 1.378893,$$

whereas LoRA-v2 obtains

$$4.371395 \pm 0.766093.$$

Thus, after rule-constrained fine-tuning, the preference-shift score increases by approximately 2.981 points.

Because the metric measures the change in relative preference between the forbidden action and the strongest legal alternative, this increase indicates that the fine-tuned model responds to the introduction of a prohibition by more strongly reallocating preference toward a legal alternative.

The smaller standard deviation observed for LoRA-v2 also indicates that the strengthened constraint response is more consistent across the evaluated probes.

Importantly, this result should not be interpreted merely as stronger suppression of forbidden actions. The metric explicitly incorporates the strongest legal alternative and therefore captures a change in the relative organization of candidate-action preferences.

5.2.2 Scenario-Level Results

The 60 probes are divided equally across six controlled scenarios, with 10 probes per scenario. Table 5 reports the corresponding mean scores.

LoRA-v2 achieves a higher constraint-sensitive score in five of the six scenarios.

The largest improvements occur in `pair_88`, `rocket_BR`, and `real_pair_QQ_landlord`. These scenarios exhibit Base-to-LoRA increases of 4.975, 4.312, and 3.952, respectively.

The `straight_34567` scenario differs from the others. The Base model already exhibits a relatively high score of 4.168, while the LoRA-v2 score is slightly lower at 3.977.

The fine-tuning effect is therefore not uniformly positive across every individual scenario.

Nevertheless, the overall mean increases substantially, and five of the six scenario-level comparisons show clear improvement. The results support a cross-scenario enhancement of constraint-sensitive preference reallocation while also indicating that the magnitude of the effect depends on the underlying decision structure.

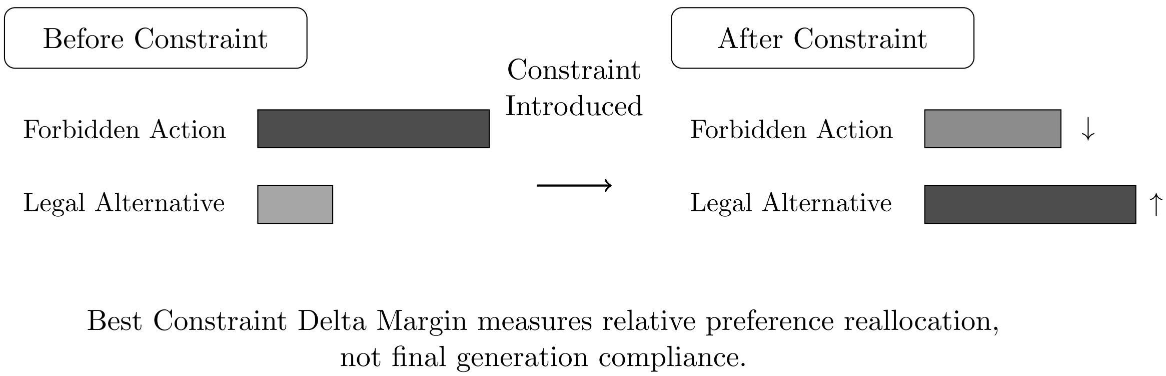


Figure 4: Illustration of constraint-sensitive preference reallocation. The best constraint delta margin measures the relative probability shift from a forbidden action toward the strongest legal alternative after an explicit constraint is introduced.

5.3 Direct Generation Diagnostic

The best constraint delta margin captures preference-level adaptation but does not directly measure final generated actions.

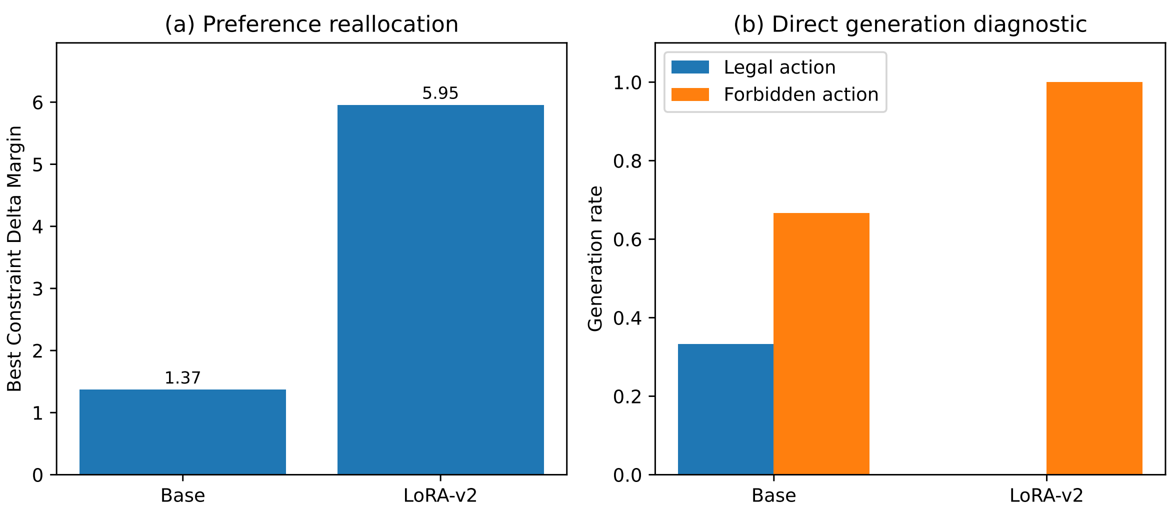


Figure 5: Comparison between preference-level adaptation and final generation behavior. The figure illustrates the difference between improved constraint-sensitive preference reallocation and final greedy decoding compliance after fine-tuning.

Under direct greedy generation, the Base model generates the forbidden action in 66.7% of the probes, while 33.3% of the generated actions correspond to legal alternatives. In contrast, LoRA-v2 generates the forbidden action in all 60 probes.

Table 6: Direct generation diagnostic on the 60 constraint-sensitive probes.

Model	Forbidden (%)	Allowed (%)	Other Legal (%)
Base	66.7	16.7	16.7
LoRA-v2	100.0	0.0	0.0

A supplementary five-probe test without the chat template produces the same qualitative outcome for LoRA-v2, with all generated first actions corresponding to the forbidden action.

These results show that the enhanced best constraint delta margin should not be interpreted as improved final-action compliance. Instead, fine-tuning strengthens the relative preference shift induced by the constraint in log-probability space, while the forbidden action can still retain the highest absolute probability during greedy decoding.

The results therefore motivate a distinction between

$$\text{constraint-sensitive preference shift} \neq \text{final generation compliance.}$$

The mechanistic analyses in the following sections concern the former quantity.

5.4 Attention-Head Causal Results

The behavioral results establish that LoRA-v2 exhibits substantially stronger constraint-sensitive preference reallocation than the Base model. We next examine whether this behavioral difference is accompanied by changes in the functional importance of individual attention heads.

The causal analysis is performed on Qwen2.5-7B.

5.4.1 Exploratory Attention-Based Screening

Before performing the complete intervention scan, we explored whether constraint-related attention heads could be identified through a three-stage attention-based screening procedure consisting of targeting, correlation, and counterfactual filtering.

For the Base model, the screening process produced

$$784 \to 157 \to 22 \to 14$$

candidate heads.

In contrast, the standard LoRA-v2 screening produced

$$784 \to 157 \to 0 \to 0.$$

A relaxed LoRA-v2 setting using 100 samples increased the number of heads surviving the targeting stage to 235, but no heads survived the subsequent correlation stage:

$$784 \to 235 \to 0 \to 0.$$

These results indicate that simple attention–behavior correlation does not provide a stable basis for identifying constraint-relevant components after fine-tuning.

Rather than treating the absence of correlated heads as evidence that the model lacks localized functional dependence, we therefore proceed to direct causal intervention.

5.4.2 Complete Single-Head Causal Scan

Qwen2.5-7B contains 28 Transformer layers and 28 attention heads per layer, resulting in 784 attention heads.

Every attention head is independently ablated in both the Base and LoRA-v2 models, yielding

$$784 \times 2 = 1568$$

single-head interventions.

For each head, we compute the causal contribution defined in Section 3.4, together with the Base-to-LoRA change

$$\Delta CE(l,h) = CE_{\text{LoRA}}(l,h) - CE_{\text{Base}}(l,h).$$

The full scan reveals a highly heterogeneous distribution.

Most attention heads show relatively small Base-to-LoRA changes, while a much smaller subset exhibits substantially larger positive ΔCE values. Negative values are also observed, indicating that some heads become less supportive of the measured behavior after fine-tuning or that their removal improves the preference-shift score.

Figure 6 summarizes the distribution of positive causal contribution across Transformer layers.

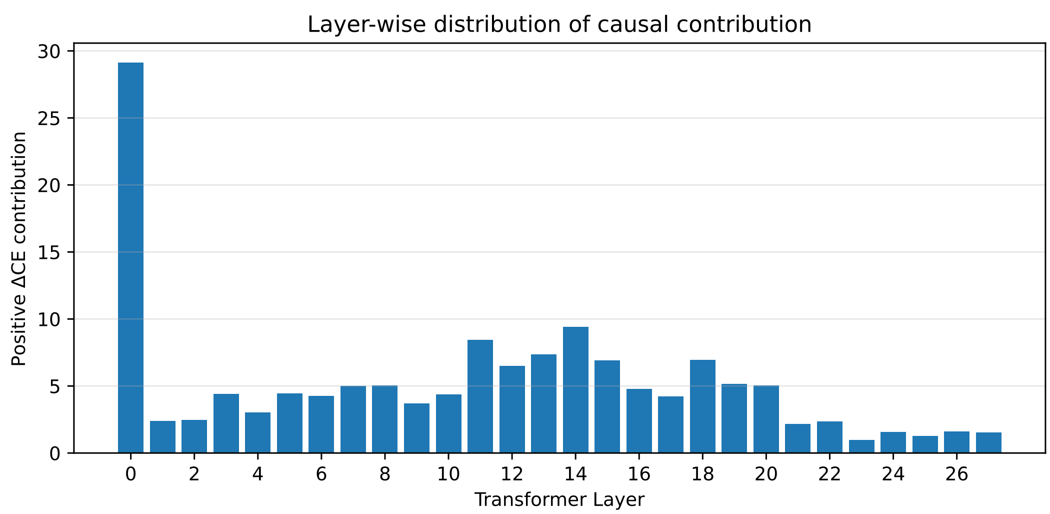


Figure 6: Layer-wise distribution of positive causal contribution measured by ΔCE across Qwen2.5-7B attention heads. The distribution shows non-uniform changes in causal contribution across Transformer layers after fine-tuning.

The causal changes are therefore not distributed uniformly across the attention system.

Table 7 reports the ten heads with the largest positive ΔCE values.

Several of the strongest causal changes are concentrated in Layer 0. In particular, five of the ten highest-ranked heads are located in the first Transformer layer.

However, high positive ΔCE values are not restricted to the earliest layer. Heads in Layers 10, 11, 18, 19, and 20 also appear among the ten highest-ranked components.

The mechanism therefore cannot be characterized as limited to early layers. Rather, the strongest individual causal amplification is concentrated in several early heads while additional high-impact components remain distributed across intermediate layers.

Table 7: Top ten Qwen2.5-7B attention heads ranked by Base-to-LoRA causal change ΔCE.

Rank	Attention Head	ΔCE
1	L0-H22	7.062
2	L0-H24	6.671
3	L0-H26	4.552
4	L0-H23	2.996
5	L0-H15	2.486
6	L20-H21	2.101
7	L18-H18	2.055
8	L11-H22	2.046
9	L10-H14	1.746
10	L19-H0	1.595

5.4.3 Representative Head: L0-H22

L0-H22 provides the clearest example of causal amplification after fine-tuning.

For the Base model, the original constraint-sensitive score is

$$S_{\text{Base}} = 1.390711.$$

After ablating L0-H22, the score decreases to

$$S_{\text{Base,ablated}} = 0.751787.$$

The corresponding causal effect is therefore

$$CE_{\text{Base}}(\text{L0-H22}) = 1.390711 - 0.751787 = 0.638924.$$

Thus, L0-H22 already makes a positive contribution to the measured behavior in the Base model.

For LoRA-v2, the original score is

$$S_{\text{LoRA}} = 4.371395.$$

After ablating the same head, the score becomes

$$S_{\text{LoRA,ablated}} = -3.329531.$$

The resulting causal effect is

$$CE_{\text{LoRA}}(\text{L0-H22}) = 4.371395 - (-3.329531) = 7.700926.$$

Therefore,

$$\Delta CE(\text{L0-H22}) = 7.700926 - 0.638924 = 7.062002.$$

This result is important because the head is not absent or functionally irrelevant in the Base model.

Instead, its contribution is positive before fine-tuning and becomes dramatically larger afterward.

This pattern is consistent with selective amplification or functional reconfiguration of an existing computational component rather than the simple appearance of an entirely new attention-head function.

5.4.4 Causal Heterogeneity Across Heads

The complete scan reveals several qualitatively different patterns of causal change.

For some heads,

$$CE_{\text{LoRA}} \gg CE_{\text{Base}} > 0,$$

indicating strong amplification of an already positive Base-model contribution.

For other heads, the Base contribution is weak while the LoRA contribution becomes substantially positive.

A large number of heads exhibit

$$\Delta CE \approx 0,$$

indicating comparatively stable functional importance before and after fine-tuning.

Finally, some heads exhibit

$$\Delta CE < 0,$$

showing that fine-tuning can also reduce the measured contribution of individual components.

These results support a sparse and non-uniform redistribution of causal dependence rather than uniform strengthening of attention heads across the network.

They also explain why attention-based correlation alone is insufficient for identifying the relevant components: strong causal importance after fine-tuning does not necessarily correspond to a simple observable attention–behavior correlation.

5.4.5 Top-k Joint-Ablation Validation

Single-head causal effects identify components whose individual importance changes after fine-tuning, but they do not establish whether the highest-ranked heads form a collectively important functional subset.

We therefore jointly ablate the Top-k heads ranked by ΔCE for

$$k \in \{5, 10, 27\},$$

and compare the resulting behavioral degradation against Global Random and Matched Random controls, each averaged over five independently sampled control sets.

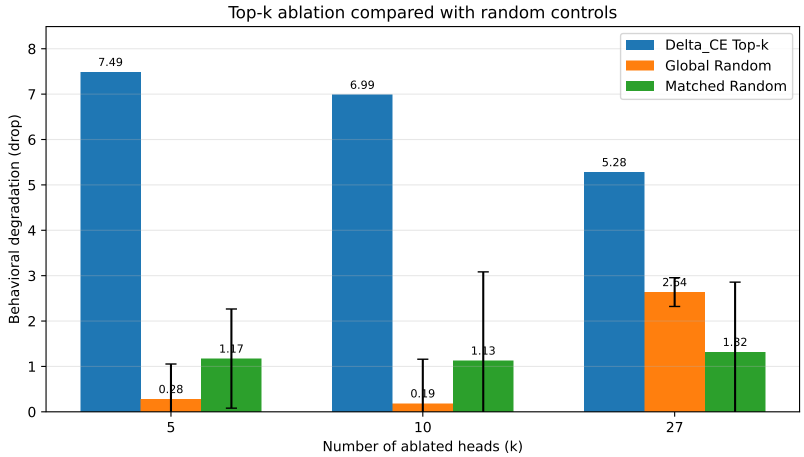
The causal ablation comparison is summarized in Figure 7.

Across all three intervention sizes, Top-k ablation causes greater behavioral degradation than either random control.

At $k = 5$, removing the five highest-ranked heads produces a drop of 7.489, compared with 0.281 under Global Random and 1.173 under Matched Random.

At $k = 10$, the Top-k drop remains large at 6.987, whereas Global Random and Matched Random produce drops of only 0.186 and 1.130, respectively.

At $k = 27$, Top-k ablation produces a drop of 5.281. Although the Global Random effect becomes larger at this intervention size, reaching 2.638, it remains substantially below the Top-k effect. The corresponding Matched Random drop is 1.320.



k	Top-k	Global Random	Matched Random
5	7.489	0.281	1.173
10	6.987	0.186	1.130
27	5.281	2.638	1.320

Figure 7: Causal validation of Delta$_{CE}$-selected attention heads. Top-k ablation produces substantially larger behavioral degradation than Global Random and Matched Random controls, indicating that the identified heads form a functionally distinctive subset rather than arbitrary or layer-dependent components. Error bars indicate standard deviation across five random control runs.

These results provide stronger evidence that the ΔCE ranking identifies a functionally distinctive subset of attention heads.

The Global Random comparison shows that the result cannot be explained simply by removing the same number of arbitrary attention heads.

The Matched Random comparison provides a stricter control by preserving the layer distribution of the Top-k set. The persistence of a larger Top-k effect under this control indicates that the result also cannot be explained solely by the network depths at which the selected heads are located.

5.4.6 Non-Monotonic Joint-Ablation Effects

An important feature of the Top-k results is that behavioral degradation does not increase monotonically with k.

The Top-k drops are

$$7.489 \to 6.987 \to 5.281$$

for

$$k = 5,\ 10,\ 27,$$

respectively.

Thus, removing additional high-ranked heads does not lead to progressively larger degradation.

This finding indicates that individual attention-head causal effects should not be interpreted as independent additive quantities.

In general,

$$CE(h_1, \ldots, h_k) \neq \sum_{i=1}^{k} CE(h_i).$$

The non-monotonic pattern is consistent with the possibility of redundancy, compensation, inhibitory interactions, or other nonlinear dependencies among attention heads.

The present experiment does not distinguish among these possibilities. Nevertheless, it shows that a relatively small set of highly ranked heads can have a particularly strong joint effect and that extending the intervention to a larger set changes the functional interaction among the remaining components.

5.4.7 Summary of the Causal Results

The causal experiments provide four main findings.

First, attention-based correlation screening and direct causal intervention yield different conclusions, emphasizing that observable attention patterns do not necessarily identify functionally important components.

Second, the complete 784-head scan reveals a sparse and heterogeneous redistribution of causal importance after fine-tuning rather than a uniform change across the attention system.

Third, representative heads such as L0-H22 already contribute positively in the Base model but become substantially more important after fine-tuning, supporting an interpretation based on selective amplification or reconfiguration of pre-existing mechanisms.

Finally, Top-k joint ablation demonstrates that the heads with the largest positive ΔCE values form a functionally meaningful subset. Their removal produces substantially greater behavioral degradation than both globally random and layer-matched random controls across all tested intervention sizes.

These causal results directly concern the best constraint delta margin. They establish that the learned constraint-sensitive preference reallocation depends selectively on a subset of attention heads whose functional contribution is strengthened after fine-tuning.

They do not, by themselves, establish that the same heads causally mediate the cross-task ACL Bench improvements.

6 Analysis and Discussion

The experiments reveal a consistent but non-uniform picture of the behavioral and mechanistic changes observed after fine-tuning on the rule-constrained mixed corpus. At the behavioral level, fine-tuned models achieve higher overall performance on ACL Bench across all three evaluated model families, while the category-level changes vary substantially across models. At the source-task level, Qwen2.5-7B exhibits a substantially stronger shift from forbidden actions toward legal alternatives under explicit constraints. At the mechanistic level, the complete attention-head intervention scan shows that this behavioral change is accompanied by a sparse redistribution of causal contribution, and joint Top-k ablation further confirms that the highest-ranked heads form a functionally distinctive subset.

Taken together, these results suggest that fine-tuning should not be interpreted simply as adding a uniformly stronger reasoning capability. Instead, the observed changes are selective: particular behavioral operations are strengthened, different base models express these changes differently, and the internal dependence supporting constraint-sensitive preference shift becomes concentrated on a limited subset of attention heads.

6.1 Selective Rather Than Uniform Task Transfer

The ACL Bench results show overall improvement after LoRA-v2 fine-tuning for DeepSeek-R1-Distill-Llama-8B, Qwen2.5-7B, and Llama3.1-8B. However, the category-level results demonstrate that this improvement is not distributed uniformly across the three reasoning categories.

For Qwen2.5-7B, the overall gain is primarily driven by large improvements in Hierarchy Logic and Pattern Recognition, while Counter Logic decreases. Llama3.1-8B shows a particularly large gain in Pattern Recognition but only limited changes in the other two categories. DeepSeek-R1-Distill-Llama-8B exhibits smaller but more broadly distributed improvements.

These different transfer profiles argue against interpreting the fine-tuning effect as a general increase in reasoning ability. If the intervention simply improved reasoning in a task-independent manner, more consistent category-level gains would be expected.

A more appropriate interpretation is that fine-tuning strengthens certain operations that overlap with the structural requirements of the target tasks. Such operations may include ordered comparison, structured pattern processing, conditional candidate selection, and adjustment of action preference under changing rules.

The resulting transfer can therefore be described as *selective structural transfer*. In this view, the relevant unit of transfer is not necessarily the source task itself, but reusable operations repeatedly exercised within that task.

Conceptually,

$$\begin{aligned} &\text{constraint introduced} \\ &\rightarrow \text{preferred action prohibited} \\ &\rightarrow \text{preference shifted toward a legal alternative.} \end{aligned}$$

This interpretation also explains why different base models can exhibit different transfer profiles under the same general fine-tuning framework.

6.2 The Base Model Shapes the Transfer Outcome

The cross-model results indicate that the effect of fine-tuning depends strongly on the starting model.

The same rule-constrained training paradigm does not produce an identical pattern of downstream improvement across Qwen2.5-7B, Llama3.1-8B, and DeepSeek-R1-Distill-Llama-8B. Instead, each model family exhibits its own combination of gains, stability, and degradation across ACL categories.

This suggests that task transfer can be viewed as an interaction among three factors:

$$\text{Transfer} = f(\text{Training Structure}, \text{Base Model}, \text{Target Structure}).$$

The fine-tuning data determine which behavioral operations are repeatedly reinforced, but the base model determines how those operations are represented and how easily they can be adapted. The target task then determines whether the modified representations or decision procedures are useful.

This perspective has two implications.

First, transfer should not be inferred from a single model family. A result that appears broad in one model may be highly selective in another.

Second, negative transfer in a specific category does not necessarily contradict the existence of meaningful transfer elsewhere. The Qwen2.5-7B Counter Logic decrease, for example, occurs alongside large improvements in Hierarchy Logic and Pattern Recognition.

The fine-tuning effect is therefore better understood as a redistribution of task competence rather than a simple scalar increase in capability.

6.3 Constraint Learning as Preference Reallocation

The constraint-sensitive probe results clarify what type of behavioral adaptation is learned by LoRA-v2.

A naive measure of rule following might examine only whether the probability of a prohibited action decreases after a constraint is introduced. However, suppression alone does not guarantee successful decision adaptation. Probability mass may move toward another invalid, irrelevant, or low-quality action.

The best constraint delta margin instead measures the change in relative preference between the forbidden action and the strongest legal alternative. This allows the learned behavior to be interpreted as *preference reallocation*.

The relevant behavioral pattern is therefore

$$\begin{aligned}\text{constraint introduced} &\rightarrow \text{preferred action prohibited}\\ &\rightarrow \text{preference shifted toward a legal alternative.}\end{aligned}$$

The substantially larger mean score for LoRA-v2 indicates that the fine-tuned model is more responsive to changes in the admissible action space.

This distinction is important because the constraint-injected examples differ from ordinary supervised examples. Standard supervision often reinforces a mapping of the form

$$x \rightarrow a^*,$$

whereas the special examples expose the model to

$$(x, c) \rightarrow a^c,$$

where the additional condition changes the appropriate output.

The model is therefore repeatedly trained on situations in which the correct decision depends not only on the underlying state but also on an additional rule.

This form of supervision may encourage the use of conditional decision policies rather than fixed state–action mappings.

However, the present results do not yet isolate the incremental effect of the constraint-injected examples from the broader effect of domain fine-tuning. A strictly matched LoRA-v1 versus LoRA-v2 experiment would be required to establish that distinction directly.

6.4 Fine-Tuning Redistributes Attention-Head Causal Dependence

The complete single-head intervention analysis shows that the stronger constraint-sensitive preference shift is accompanied by substantial changes in the causal importance of attention heads.

These changes are highly heterogeneous.

Most heads exhibit relatively small Base-to-LoRA changes, while a much smaller subset shows large positive ΔCE values. Other heads exhibit negative changes.

This pattern is inconsistent with a uniform strengthening of the entire attention system.

Instead, the results support a *functional reconfiguration* interpretation: fine-tuning changes which pre-existing components the model depends on for the measured behavior.

The representative L0-H22 result is particularly informative. This head already contributes positively to constraint-sensitive preference shift in the Base model, but its causal effect becomes dramatically larger after fine-tuning.

This pattern is consistent with prior findings that fine-tuning can enhance mechanisms already present in a pretrained model rather than necessarily constructing entirely new computational structures [11].

The current results extend this idea to rule-constrained decision behavior. At least for some heads, fine-tuning appears to amplify an existing functional contribution.

At the same time, the complete causal map includes multiple forms of change:

$$CE_{\text{LoRA}} \gg CE_{\text{Base}} > 0,$$

for strongly amplified heads;

$$\Delta CE \approx 0,$$

for comparatively stable heads; and

$$\Delta CE < 0,$$

for heads whose contribution becomes weaker after fine-tuning.

The resulting mechanism is therefore better characterized as selective strengthening, weakening, and redistribution rather than simple amplification.

6.5 Top-k Validation Strengthens the Causal Interpretation

Single-head ranking alone leaves open the possibility that high ΔCE values reflect isolated intervention effects that do not identify a meaningful functional subset.

The Top-k joint-ablation experiments provide stronger evidence against this interpretation.

Across

$$k \in \{5, 10, 27\},$$

removing the highest-ranked heads produces greater behavioral degradation than both Global Random and Matched Random controls.

Both random-control conditions are averaged across five independently sampled control sets. The Global Random comparison therefore tests whether removing an arbitrary set of the same size is sufficient to explain the degradation, while Matched Random additionally controls for the layer distribution of the selected heads.

The persistence of a substantially larger Top-k effect under both controls supports the conclusion that the identity of the selected heads matters.

In particular, the Matched Random result is important because several of the highest-ranked heads occur in early layers. Without a layer-matched comparison, one alternative explanation would be that intervention in these layers is simply more disruptive in general.

The control results weaken this explanation.

The current evidence therefore supports the interpretation that the high-ΔCE heads form a selectively important functional subset for constraint-sensitive preference shift.

The $k = 27$ setting provides an additional threshold-based view of this subset, because it corresponds to the complete set of heads satisfying

$$\Delta CE > 1.$$

Thus, the joint-ablation results validate both small high-ranking subsets and a broader set selected by a causal-change threshold.

6.6 Attention-Head Contributions Are Non-Additive

Although Top-k ablation consistently produces larger degradation than the random controls, the effect does not increase monotonically with k.

The behavioral degradation decreases from Top-5 to Top-10 and further to Top-27.

This result is important because it shows that individual head contributions cannot be interpreted as independent additive quantities.

In general,

$$CE(h_1, \ldots, h_k) \neq \sum_{i=1}^{k} CE(h_i).$$

Several mechanisms could produce this pattern.

First, multiple heads may perform partially redundant functions. Once one influential subset has been removed, ablating additional heads may contribute relatively little additional damage.

Second, compensatory interactions may allow remaining components to reorganize after a larger intervention.

Third, some heads may exert partially opposing or inhibitory effects on the measured behavior. Removing such heads together with strongly positive heads could reduce the net degradation.

Finally, larger interventions may alter the network state sufficiently that individual causal effects measured under single-head ablation no longer predict their joint effect.

The present experiments do not distinguish among these possibilities.

Nevertheless, the non-monotonic result provides evidence that the mechanism is interaction-dependent rather than a simple collection of independently contributing heads.

This also suggests that future mechanistic analysis should move beyond ranking isolated components toward explicit analysis of interactions among selected heads.

6.7 Attention Correlation Is Not Equivalent to Causal Importance

The exploratory attention-based screening provides an instructive contrast with the causal intervention results.

The Base model produced a small set of heads that survived targeting, correlation, and counterfactual screening. In contrast, no LoRA-v2 heads survived the correlation stage under either the standard or relaxed screening setting.

Taken alone, this observation could be misinterpreted as evidence that the fine-tuned model lacks localized attention components related to the constraint behavior.

The causal intervention results show that this interpretation would be incorrect.

Several LoRA-v2 attention heads have large behavioral effects when ablated, and the Top-k subset produces substantially greater degradation than both random controls.

The discrepancy therefore reinforces the distinction

$$\text{attention correlation} \neq \text{causal contribution}.$$

Observable attention patterns do not necessarily provide a reliable measure of functional dependence.

An attention head may participate in a distributed computation without exhibiting a simple monotonic relationship between attention weight and behavioral output.

For this reason, the present study treats attention-based screening as exploratory and causal intervention as the primary mechanistic evidence.

6.8 Early-Layer Amplification and Distributed Processing

Several of the strongest positive ΔCE values occur in Layer 0, suggesting that early attention processing may become particularly important after fine-tuning.

One possible interpretation is that these heads influence the initial representation of information relevant to candidate actions or explicit constraints before that information is propagated through deeper layers.

However, the top-ranked set is not restricted to Layer 0.

High positive causal changes also occur in Layers 10, 11, 18, 19, and 20.

The mechanism should therefore not be described as purely early-layer localized.

A more cautious interpretation is that the strongest individual amplification occurs in several early heads, while constraint-sensitive computation remains distributed across multiple depths.

The layer-matched random control further indicates that the Top-k effect cannot be reduced to a simple statement that "early-layer ablation is generally more disruptive."

Rather, particular heads within those layers exhibit selectively increased functional importance.

6.9 Relationship Between Constraint Behavior and ACL Transfer

The current study provides three complementary levels of evidence:

$$\text{ACL Bench} \rightarrow \text{cross-task behavioral transfer,}$$

$$\text{constraint probes} \rightarrow \text{stronger constraint-sensitive preference reallocation,}$$

and

$$\text{head ablation} \rightarrow \text{selective causal dependence on an amplified head subset.}$$

These results form a coherent picture, but an important causal distinction must be maintained.

The attention-head interventions are evaluated using the best constraint delta margin. They therefore directly establish that the identified heads contribute to the learned constraint-sensitive preference reallocation.

They do not directly establish that the same heads are responsible for the ACL Bench improvements.

At present, the evidence supports

$$\begin{aligned}&\text{rule-constrained fine-tuning}\\&\rightarrow \text{stronger constraint-sensitive preference shift}\\&\rightarrow \text{enhanced causal contribution of selected attention heads.}\end{aligned}$$

while independently showing

$$\text{rule-constrained fine-tuning} \rightarrow \text{selective ACL task transfer.}$$

A direct mechanistic bridge between these two paths would require evaluating ACL Bench under Top-k and matched-random ablation.

If ACL performance decreases more strongly under Top-k intervention than under the matched control, this would support the hypothesis that some of the same internal components contribute to both source-domain constraint adaptation and cross-task transfer.

Conversely, if ACL performance remains comparatively stable while the preference-shift score collapses, this would suggest that the two effects rely on partially distinct mechanisms.

Both outcomes would be informative.

6.10 Broader Implications

The results have several implications for the study of fine-tuned language models.

First, domain-specific fine-tuning does not necessarily produce only domain-specific behavioral changes. If a source task repeatedly exercises reusable operations, some of those operations may transfer to tasks with different surface forms.

Second, overall benchmark accuracy is insufficient for characterizing transfer. Category-level analysis reveals capability trade-offs that would otherwise remain hidden.

Third, fine-tuning changes not only model outputs but also the distribution of internal functional dependence. Two models with the same underlying architecture can rely on substantially different components after adaptation.

Fourth, causal changes can be sparse. The large Top-k effects relative to random controls indicate that a limited subset of heads can become disproportionately important.

Finally, mechanistic effects are interaction-dependent. The non-monotonic Top-k result cautions against treating component rankings as additive decompositions of behavior.

Together, these observations support a view of fine-tuning as both a behavioral and mechanistic reorganization process.

6.11 Limitations

Several limitations remain.

Single source domain. The present study uses Dou Dizhu as the primary structured decision environment. Although the domain contains rich hierarchical, pattern-based, and constraint-dependent operations, additional source domains are required to determine how broadly the observed transfer patterns generalize.

Self-constructed transfer benchmark. ACL Bench is intentionally designed to capture operations structurally related to the source domain. This makes it suitable for testing the proposed transfer hypothesis, but the observed improvements should not be interpreted as broad general reasoning enhancement. Evaluation on additional independent benchmarks would provide stronger external validation.

Constraint-specific contribution is not yet fully isolated. The current main results compare Base and LoRA-v2 models. A strictly matched LoRA-v1 versus LoRA-v2 experiment is still needed to isolate the incremental effect of the constraint-injected examples from ordinary domain fine-tuning.

Limited training-run replication. The current results are primarily based on individual fine-tuned checkpoints rather than multiple independently trained random seeds. Repeated fine-tuning runs would provide stronger evidence for across-run stability.

Mechanistic analysis is concentrated on one model family. The complete 784-head scan and Top-k causal validation are currently performed on Qwen2.5-7B. It remains unclear whether other model families exhibit similar causal redistribution.

Top-k ablation does not identify a complete circuit. Although joint ablation confirms that the highest-ranked heads form a functionally important subset, the non-monotonic effects indicate unresolved redundancy, compensation, or nonlinear interactions. The selected heads should therefore not be interpreted as a complete isolated circuit.

No direct causal test of ACL transfer. The identified heads are causally validated for constraint-sensitive preference shift but have not yet been tested directly on ACL Bench under ablation. The present mechanistic evidence therefore explains the learned constraint-sensitive preference reallocation rather than directly establishing the mechanism of cross-task transfer.

6.12 Future Directions

The most immediate extension is a strictly matched LoRA-v1 versus LoRA-v2 comparison under identical training and evaluation settings. This experiment would more directly isolate the incremental contribution of the constraint-injected examples.

A second priority is cross-task causal evaluation. Applying the Top-k and Matched Random interventions during ACL Bench evaluation would test whether the attention heads supporting constraint-sensitive preference shift also contribute to the observed transfer effect.

A third direction is cross-model mechanistic analysis. Repeating either the complete head scan or a reduced intervention analysis on additional model families would help determine whether the observed causal redistribution is architecture-specific or general.

Finally, interaction-level analysis among the highest-ranked heads may help explain the non-monotonic Top-k effects and distinguish redundancy, compensation, and inhibitory interactions.

6.13 Discussion Summary

The current results support a multi-level interpretation of rule-constrained fine-tuning.

At the behavioral level, fine-tuning is associated with selective cross-task transfer rather than uniform enhancement of general reasoning.

At the source-task level, the fine-tuned model exhibits substantially stronger preference reallocation when an otherwise preferred action becomes prohibited.

At the mechanistic level, this behavioral adaptation is accompanied by a sparse and heterogeneous redistribution of attention-head causal contribution.

Joint Top-k ablation further shows that the highest-ranked heads form a selectively important functional subset: their removal produces greater behavioral degradation than both globally random and layer-matched random controls.

At the same time, the non-monotonic joint-ablation effects demonstrate that attention-head contributions are not simply additive.

The resulting picture can be summarized as

$$\begin{aligned}&\text{structured rule-constrained fine-tuning}\\&\rightarrow \text{conditional behavioral adaptation}\\&\rightarrow \text{selective cross-task transfer.}\end{aligned}$$

together with

$$\text{conditional behavioral adaptation} \rightarrow \text{sparse redistribution of internal causal dependence.}$$

The remaining major question is whether these two paths share the same causal components. Establishing that connection would provide a more complete account linking training intervention, behavioral transfer, and internal mechanism.

7 Conclusion

This study investigates task transfer, constraint-sensitive preference reallocation, and internal causal dependence after fine-tuning on a rule-constrained mixed corpus. Using Dou Dizhu as a structured rule-based source domain, we construct a mixed fine-tuning corpus that includes explicit prohibitive constraints requiring the model to revise its preferred action and select a legal alternative.

To evaluate cross-task transfer, we introduce ACL Bench, a 300-item benchmark covering Hierarchy Logic, Pattern Recognition, and Counter Logic. Across DeepSeek-R1-Distill-Llama-8B, Qwen2.5-7B, and Llama3.1-8B, all models achieve higher overall ACL Bench accuracy after fine-tuning on the rule-constrained mixed corpus, while the category-level effects differ substantially across model families. These results indicate that the observed transfer is selective and model-dependent rather than a uniform improvement in general reasoning ability.

We further evaluate constraint-sensitive preference reallocation using controlled decision probes and the best constraint delta margin. The fine-tuned Qwen2.5-7B model exhibits a substantially stronger shift in relative preference from prohibited actions toward legal alternatives, showing that fine-tuning affects not only action suppression but also the reallocation of decision preference under changing rules.

At the mechanistic level, complete single-head causal ablation reveals a sparse and heterogeneous pattern of changes in attention-head causal contribution after fine-tuning. A limited subset of heads exhibits substantially amplified causal contribution to constraint-sensitive preference shift. Joint Top-k ablation further shows that removing the highest-ranked heads causes greater behavioral degradation than both globally random and layer-matched random controls across all tested intervention sizes. These results provide causal evidence that a functionally important subset of attention heads contributes to the learned constraint-sensitive preference shift rather than reflecting uniform changes across the attention system.

The non-monotonic Top-k effects further indicate that attention-head contributions are not simply additive and may involve redundancy, compensation, or nonlinear interaction.

Several questions remain open. The current main results do not yet isolate the incremental contribution of the constraint-injected examples through a strictly matched LoRA-v1 versus LoRA-v2 comparison, and the identified attention heads have not yet been causally tested on ACL Bench itself. Therefore, the present mechanistic results directly explain constraint-sensitive decision behavior, while the relationship between these causal components and cross-task transfer remains to be established.

Overall, this work provides a unified experimental framework for studying fine-tuning at both the behavioral and mechanistic levels. By combining structured source-domain adaptation, cross-task transfer evaluation, constraint-sensitive preference-shift measurement, and causal attention-head intervention, the study provides evidence that fine-tuning can induce selective capability transfer while selectively altering the causal contributions of internal components associated with constraint-sensitive behavior.

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